

**Assignment No.: 2**

**Title: 4 bit BCD Adder**

**Course Outcome:**

**CO201C.1: Construct Combinational Circuit for Digital Applications**

**Date of Completion:**

**Assessment Grade / Marks:**

**Assessor's Sign with Date:**

## Assignment No. 2

**Title:** BCD Adder

**Objective:** To learn working of BCD adder using IC 7483.

**Problem Statement:** Design and implement 4-bit BCD adder using IC 7483.

**Course Outcome:** CO201C.1

**Blooms Taxonomy Category:** Applying

**Hardware Requirements:** --

1. Digital Trainer Kit and Patch Cords
2. IC 7408(Quad 2 input AND gate)
3. IC 7432(Quad 2 input OR gate)
4. IC 7483 (BCD Adder)

**Theory:**

- **BCD Adder:**

BCD stands for binary coded decimal. It is used to perform the addition of BCD numbers. A BCD digit can have any of ten possible four-bit representations. Suppose, we have two 4-bit numbers A and B. The value of A and B can vary from 0(0000 in binary) to 9(1001 in binary) because we are considering decimal numbers.

A BCD adder is a circuit for the addition of two binary-coded decimal numbers. BCD is another format used in representing numbers where each digit will be represented using a 4-bit binary code. In BCD, it is obvious that the most significant bit (leftmost) is always 0, and therefore each digit will be confined to the range 0-9.

- **Three cases of BCD adder with examples**

**Case I: Sum Equals 9 or less with carry 0**

Let us consider additions of 3 and 6 in BCD.

|       |       |   |   |   |             |
|-------|-------|---|---|---|-------------|
| 6     | 0     | 1 | 1 | 0 | ← BCD for 6 |
| + 3   | 0     | 0 | 1 | 1 | ← BCD for 3 |
| <hr/> | <hr/> |   |   |   |             |
| 9     | 1     | 0 | 0 | 1 | ← BCD for 9 |

The addition is carried out as in normal binary addition and the sum is 1 0 0 1, which is BCD code for 9.

### Case II: Sum greater than 9 with carry 0

Let us consider addition of 6 and 8 in BCD

|     |         |                      |
|-----|---------|----------------------|
| 6   | 0 1 1 0 | ← BCD for 6          |
| + 8 | 1 0 0 0 | ← BCD for 8          |
| 14  | 1 1 1 0 | ← Invalid BCD number |

The sum 1 1 1 0 is an invalid BCD number. This has occurred because the sum of the two digits exceeds 9. Whenever this occurs the sum has to be corrected by the addition of six (0110) in the invalid BCD number, as shown below

|                                                                                |                                                                                |                        |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------|
| 6                                                                              | 0 1 1 0                                                                        | ← BCD for 6            |
| + 8                                                                            | 1 0 0 0                                                                        | ← BCD for 8            |
| 14                                                                             | 1 1 1 0                                                                        | ← Invalid BCD number   |
|                                                                                | + 0 1 1 0                                                                      | ← Add 6 for correction |
|                                                                                | 0 1 0 0                                                                        |                        |
| 0 0 0 1                                                                        | 0 1 0 0                                                                        |                        |
| <div style="border-top: 1px solid black; width: 100px; margin: 0 auto;"></div> | <div style="border-top: 1px solid black; width: 100px; margin: 0 auto;"></div> | ← BCD for 14           |

After addition of 6 carry is produced into the second decimal position.

### Case III: Sum equals 9 or less with carry 1

Let us consider addition of 8 and 9 in BCD

|     |                 |                        |
|-----|-----------------|------------------------|
| 8   | 1 0 0 0         | ← BCD for 8            |
| + 9 | 1 0 0 1         | ← BCD for 9            |
| 17  | 0 0 0 1 0 0 0 1 | ← Incorrect BCD result |

In this, case, result (0001 0001) is valid BCD number, but it is incorrect. To get the correct BCD result correction factor of 6 has to be added to the least significant digit sum, as shown below

|       |   |   |   |   |   |   |           |   |   |                      |                      |
|-------|---|---|---|---|---|---|-----------|---|---|----------------------|----------------------|
| 8     |   | 1 | 0 | 0 | 0 | ← | BCD for 8 |   |   |                      |                      |
| + 9   |   | 1 | 0 | 0 | 1 | ← | BCD for 9 |   |   |                      |                      |
| <hr/> |   |   |   |   |   |   |           |   |   |                      |                      |
| 17    | 0 | 0 | 0 | 1 | 0 | 0 | 0         | 1 | ← | Incorrect BCD result |                      |
|       | + | 0 | 0 | 0 | 0 | 0 | 1         | 1 | 0 | ←                    | Add 6 for correction |
| <hr/> |   |   |   |   |   |   |           |   |   |                      |                      |
|       |   | 0 | 0 | 0 | 1 | 0 | 1         | 1 | 1 | ←                    | BCD for 17           |

Going through these three cases of BCD addition we can summarize the BCD addition procedure as follows:

1. Add two BCD numbers using ordinary binary addition.
2. If four-bit sum is equal to or less than 9, no correction is needed. The sum is in proper BCD form.
3. If the four-bit sum is greater than 9 or if a carry is generated from the four-bit sum, the sum is invalid.
4. To correct the invalid sum, add 0110<sub>2</sub> to the four-bit sum. If a carry results from this addition, add it to the next higher-order BCD digit.

Thus to implement BCD Adder Circuit we require:

- 4-bit binary adder for initial addition
- Logic circuit to detect sum greater than 9 and
- One more 4-bit adder to add 0110<sub>2</sub> in the sum if sum is greater than 9 or carry is 1.

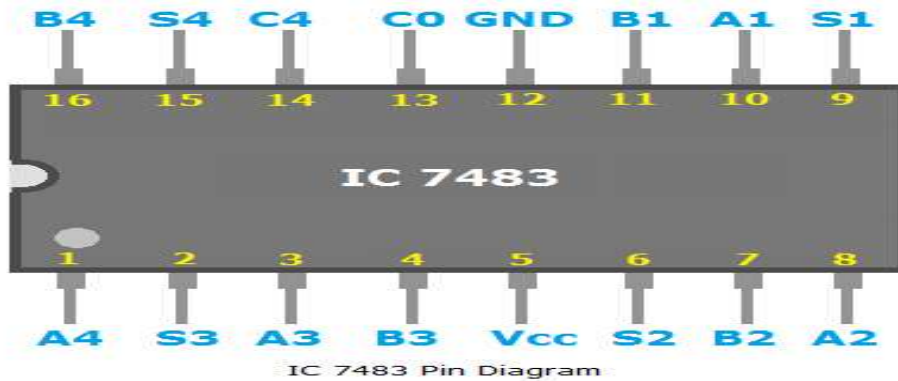
The logic circuit to detect sum greater than 9 can be determined by simplifying the Boolean expression of given BCD Adder Truth Table.

| Inputs         |                |                |                | Output |
|----------------|----------------|----------------|----------------|--------|
| S <sub>3</sub> | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | Y      |
| 0              | 0              | 0              | 0              | 0      |
| 0              | 0              | 0              | 1              | 0      |
| 0              | 0              | 1              | 0              | 0      |
| 0              | 0              | 1              | 1              | 0      |
| 0              | 1              | 0              | 0              | 0      |
| 0              | 1              | 0              | 1              | 0      |
| 0              | 1              | 1              | 0              | 0      |
| 0              | 1              | 1              | 1              | 0      |
| 1              | 0              | 0              | 0              | 0      |
| 1              | 0              | 0              | 1              | 0      |
| 1              | 0              | 1              | 0              | 1      |
| 1              | 0              | 1              | 1              | 1      |
| 1              | 1              | 0              | 0              | 1      |
| 1              | 1              | 0              | 1              | 1      |
| 1              | 1              | 1              | 0              | 1      |
| 1              | 1              | 1              | 1              | 1      |

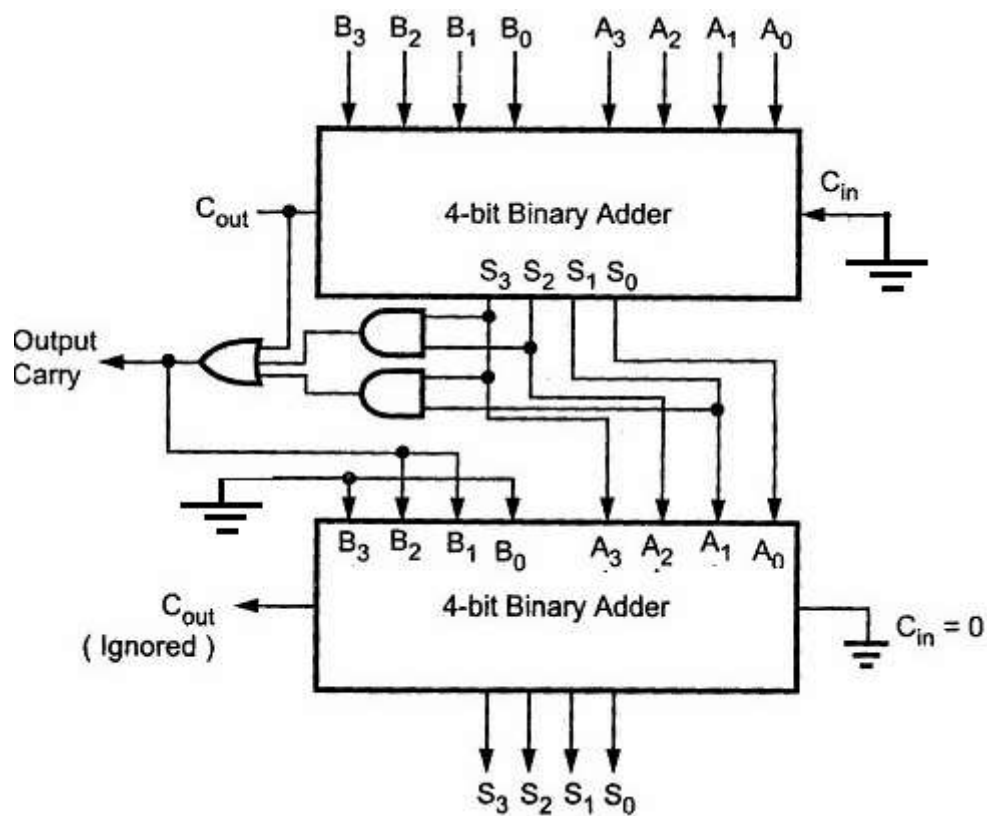
$$Y = S_3S_2 + S_3S_1$$

|                            |    |    |    |    |
|----------------------------|----|----|----|----|
| $S_3S_2 \backslash S_1S_0$ | 00 | 01 | 11 | 10 |
| 00                         | 0  | 0  | 0  | 0  |
| 01                         | 0  | 0  | 0  | 0  |
| 11                         | 1  | 1  | 1  | 1  |
| 10                         | 0  | 0  | 1  | 1  |

### IC 7483 pin configuration



- Circuit diagram of BCD adder using IC 7483



**Inference:** In this way we have implemented the BCD adder with the help of Binary adder IC 7483.